IndiaMediaLens: A Real-Time System for Monitoring Political Tone in Indian News Media

Rupsha Ray
Ashoka University
Rajiv Gandhi Education City, Haryana, India
ritwik.ray_ug2023@ashoka.edu.in

Anirban Sen
Ashoka University
Rajiv Gandhi Education City, Haryana, India
anirban.sen@ashoka.edu.in

***Abstract*—We present *IndiaMediaLens*, an interactive system for monitoring the political reporting preferences of Indian digital news sources on Google News. Rather than assigning a single label of political bias, the system analyzes article-level tone toward establishment and opposition political entities, providing both longitudinal analysis of election coverage and real-time analysis of current news coverage. IndiaMediaLens combines political entity recognition, affiliation mapping, and targeted sentiment analysis with interactive visualizations to expose reporting patterns and their underlying textual evidence. We present a detailed evaluation and discuss the system's potential as a tool for media monitoring and media literacy in India.**

***Index Terms*—Web Mining, Targeted Sentiment Analysis, Media Analysis**

I. Introduction

News media plays an important role in shaping citizens' understanding of political events, particularly in India, where a diverse news ecosystem often provides differing accounts of the same events. Yet, systematically examining these differences across outlets remains challenging. Existing computational studies of news have examined issues such as imbalances in coverage of political parties and topics [1], representation of specific issues such as caste [2], and political bias in news outlets [3]. These studies demonstrate the value of computational approaches for analyzing Indian news media, but largely focus on cross-sectional analysis of coverage imbalance and thematic representation, or subjectively defined "political bias" through human assessments. The term "bias" is inherently subjective and difficult to quantify consistently across different analytical perspectives, which makes generalization of claims around computational media studies difficult.

We propose *IndiaMediaLens*, an interactive system for examining the political reporting preferences (or tone) of Indian digital news sources, both longitudinally and for current coverage, in real time. Unlike systems that characterize reporting over a fixed period, IndiaMediaLens enables users to observe the evolution of reporting preferences over time alongside the latest coverage. Rather than assigning outlets a subjective "bias" label, the system models reporting preference as the targeted sentiment of an article toward specific political entities (parties and politicians) and their affiliations, keeping reporting preference around establishment and opposition as separate observations. This approach thus does not impose a predefined notion of pro/anti bias – both an opinion article directly supporting a political entity, and a neutrally written article that reports a positive outcome verbatim for that entity contribute to the positive tone. For example, the sentence "*The [PARTY] said it would build a million houses for the poor every year and make urban housing affordable*" contributes to positive tone toward the party despite being verbatim reporting. Thus, IndiaMediaLens exposes the underlying political reporting patterns, while leaving their interpretation to the user.

The system works as follows: the user first enables the plugin, selects whether they are interested in establishment or opposition-related tone analysis, and activates the plugin on the *Full Coverage* page of an event (accessible by clicking on "See more headlines and perspectives" under a news story) on Google News. IndiaMediaLens works in two primary phases:

- **Current tone analysis:** The system collects article URLs from the selected Google News *Full Coverage* page and retrieves their text and metadata. Named entities are identified using named entity recognition (NER) and mapped to establishment or opposition alliances using NivaDuck [4] and a pre-defined alliance mapping based on Wikipedia. A GRU based model trained on NewsMTSC news data performs targeted sentiment analysis to derive sentence-level tone toward the identified entities. The sentence- and aggregated article-level tone are visualized in real time.
- **Longitudinal tone analysis:** The same pipeline is applied offline to articles covering Indian Lok Sabha Elections from 2019 to 2024. Entity and tone annotations are precomputed and stored, and aggregated to obtain outlet-level tone. The visualization engine presents the selected outlet's overall reporting preference and its temporal evolution through interactive visualizations.

The preliminary user evaluation found that IndiaMediaLens was particularly useful for identifying and comparing reporting patterns across outlets, with participants highlighting the utility of both current and longitudinal analysis trends. To the best of our knowledge, IndiaMediaLens is the first system in India to quantify reporting preferences for 14 digital media outlets ranked on Google News, using large-scale web-based article data, in real time. By making reporting patterns and their

Mphasis AI & Applied Tech Lab at Ashoka

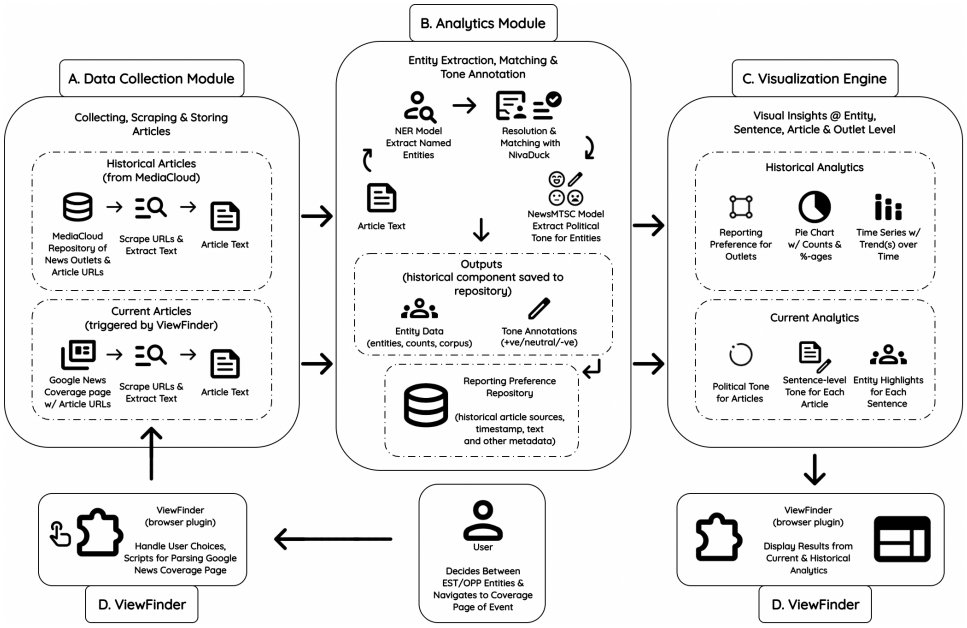


Fig. 1. IndiaMediaLens Architecture

underlying textual evidence accessible to users, the system acts as a practical media-monitoring tool – for students, journalists, policy researchers, and general news consumers – supporting media literacy, independent monitoring, and greater editorial self-reflection. While the current system relies on the coverage provided by Google News, we present it as a pilot toward systematic media monitoring in India. As one of the first systems to provide user-friendly analysis of news reporting preferences, IndiaMediaLens establishes a foundation that can be extended to incorporate additional news sources, coverage platforms, and analytical capabilities.

II. System Design

A. Data Collection Module

This module collects article URLs and extracts metadata and text for both historical and current articles. We identify digital news sources based on the Reuters Institute's Digital News Report 2025 [5], currently focusing on English-language media, with regional media left for future work. For longitudinal analysis, we focus on the *Lok Sabha Elections*, a major and recurring event that attracts extensive media coverage in India. The module uses the Media Cloud API [6], an open-source media research project. Media Cloud serves as a repository of global news to extract URLs of news articles for the selected sources using a set of relevant keywords[1] (e.g., "lok sabha elections" and "polls") for the event, alongside necessary metadata like title, author, and outlet name.

The first 200 words of each article are extracted using a web-scraping pipeline based on Newspaper3k [7] and Selenium [8], following the inverted-pyramid principle that places key information at the beginning of a story [9]. Manual analysis of 100 articles confirmed the suitability of this threshold, yielding 25,779 articles from January 2019 to December 2024 for further analysis (Figure 2).

The observed data imbalance, both across outlets and years,[2] results from data incompleteness in Media Cloud or anti-scraping measures by digital outlets. Thus, longitudinal results should be interpreted as indicative patterns, particularly for years with fewer articles. This limitation does not affect the current tone analysis, which uses articles retrieved directly from the selected Google News Full Coverage page. A similar URL, metadata, and text extraction approach is followed for the current articles for the user selected story, which appear on the Full Coverage page for an event. Unlike the offline longitudinal data collection pipeline, this pipeline functions in real-time. The module additionally decodes the URLs using Google News Decoder [10] to extract the original outlet URLs, which are masked by Google News, before proceeding to text and metadata extraction.

[1]Supplementary data available at: https://bit.ly/4wB0Khj.

[2]Temporally, we have around 7.6K articles from 2019, 1.2K annually from 2020–22, 2,379 from 2023, and 12,272 from 2024.

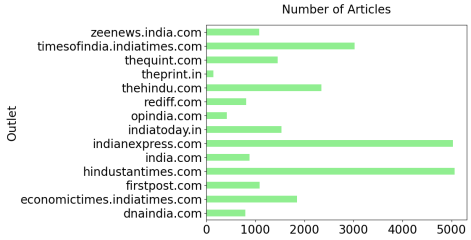


Fig. 2. Count of Articles per Outlet

B. Analytics Module

This module executes two steps to detect the reporting preferences/tone of news outlets/current articles.

(i) Entity-Alliance Mapping. We first identified whether political entities mentioned in an article belong to the establishment (EST) or opposition (OPP) alliances, including affiliated people, parties, and organizations. During our study period, the ruling coalition was led by the *Bharatiya Janata Party* and the opposition by the *Indian National Congress*. We construct a party-to-alliance mapping using Wikipedia [11], [12]. While minor changes in affiliations and alliances may introduce small inaccuracies, their impact on our findings during the study period is expected to be negligible.

To carry out entity-alliance mapping, the module uses spaCy [13] to extract sentences and named entities (using spaCy's EntityRecognizer) contained in them, from the article text. Next, it determines the political affiliation of each entity with a custom pipeline component, consisting of an X API client [14] and NivaDuck. NivaDuck provides the X handles of Indian politicians along with their political party labels. The X API client is used to retrieve the corresponding display names for these handles. These names are updated in NivaDuck. The updated data is finally joined with the party-to-alliance mapping, to form the final dataset. The module uses fuzzy matching to map each entity name extracted from the article to one of the names in the updated dataset, using a similarity threshold[3] of 95%, resulting in its mapping EST or OPP classes.

(ii) Targeted Sentiment Classification. The Analytics module uses NewsSentiment [15], a GRU-TSC model for targeted sentiment extraction (TSE) of entities. We selected this non-LLM model for its superior performance to both proprietary LLMs and open-source LLMs for TSE (see section III), and lower computational cost and runtime for large-scale analysis. Given a (sentence, entity) pair, the model predicts the entity-specific sentiment with a probability score. We retain high quality predictions with scores $\geq 75\%$, a threshold determined through manual analysis of 50 sentences. These sentences are termed *tone-bearing sentences*. For each article, we compute the mean probability for positive, neutral, and negative tone-bearing sentences, forming the article tone vector a_i.

$$a_i = \begin{bmatrix} a_i^+ & a_i^0 & a_i^- \end{bmatrix} \tag{1}$$

The aggregated tone t_i of the article is expressed as follows:

$$t_i = \underset{k \in \{+,\ 0,\ -\}}{\operatorname{argmax}} \ a_i^k \tag{2}$$

For the longitudinal tone analysis, considering an outlet N, if the number of articles with tone k is given by n_N^k, then the majority tone t_N of the outlet is given by:

$$t_N = \underset{k \in \{+,\ 0,\ -\}}{\operatorname{argmax}} \begin{bmatrix} n_N^+ & n_N^0 & n_N^- \end{bmatrix} \tag{3}$$

[3]The similarity threshold was determined after a manual analysis of 200 entity pairs.

C. Visualization Engine

This module generates and hosts GUI-based visualizations of longitudinal and current reporting-preference analytics. For the former, it uses NiceGUI [16] to provide interactive visualizations for the user selected time period: (A) a pie chart showing positive, neutral, and negative article counts, and (B) a stacked time-series chart showing tone over user-selected yearly, half-yearly, quarterly, or monthly intervals. For current articles, the side panel displays sentence-level tone with target entities and associated confidence scores. Tone is color-coded as green (positive), yellow (neutral), and red (negative).

D. ViewFinder: User Facing Browser Plugin

ViewFinder [4] is the user facing Google News based browser plugin that interacts with the remaining modules to create and present the longitudinal and current tone visualizations. Upon activation on a Google News Full Coverage page, the plugin retrieves pre-computed historical tone information and invokes the Current Article Extraction, Analytics, and Visualization modules. It adds two color-coded icons beside each article: a circular icon for the article's current tone and a square icon for the outlet's historical tone (Equation 3); with an additional gray color for current articles/outlets where the tone is 'unknown', owing to entities absent in NivaDuck or unavailability of historical data. Clicking these icons leads to the browser side panel displaying the corresponding visualizations (demonstration of the usage flow shown in Figure 3).

III. Evaluation

We manually annotated 50 opinion articles containing 361 political entities to evaluate the entity-alliance mapping exercise, obtaining an F1 score of 0.735 (precision: 0.986, recall: 0.585). The high precision indicates reliable alliance mapping. The lower recall mainly results from misses due to non-canonical entity names returned by the X API, which is less critical as prominent political entities are reliably recognized and mapped. Our future work will develop a curated dataset with formal entity names and time-varying alliance affiliations.

We evaluated TSE using 100 high-confidence NewsSentiment annotations (50 each for EST and OPP), manually labeled by two annotators with 96% inter-annotator agreement (disagreements were resolved through discussion). Compared with Llama 3.1 (8B) and GPT-5.6 Luna, NewsSentiment achieved satisfactory performance and outperformed both (Table I). While LLMs occasionally lost focus on the target entity, e.g., when sentiment differed toward EST and OPP entities in the same sentence, NewsSentiment predominantly differentiates between the two correctly. However, it struggled with colloquial expressions due to limited local context. This limitation will be addressed as part of future work, where we will fine-tune the model on Indian news data.

We conducted a preliminary qualitative evaluation (see Supplementary) with ten participants to assess the usefulness and interpretability of IndiaMediaLens. Participants performed

[4]ViewFinder plugin available at: https://github.com/Rupository/ViewFinder

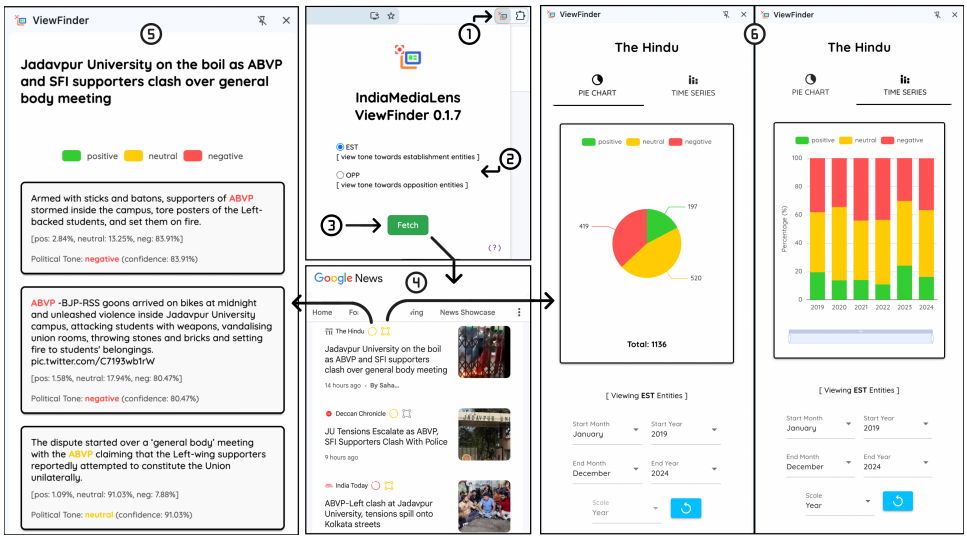


Fig. 3. ViewFinder Usage Flow: **1.** Select from Toolbar. **2.** Choose between EST/OPP Entities. **3.** Fetch Analytics. **4.** Check Aggregated Stances (circle and square icons) on the Full Coverage Page. **5.** Click Circle for Current Analysis. **6.** Click Square for Longitudinal Analysis.

TABLE I: TSE Performance: NewsSentiment vs. LLMs

	NewsSentiment	Llama 3.1 (8B)	GPT-5.6 Luna
EST	84%	54%	78%
OPP	86%	66%	74%
Overall	85%	60%	76%

tasks involving current articles and longitudinal reporting patterns, followed by Likert-scale and open-ended questions. All participants agreed or strongly agreed that the visualizations helped identify reporting patterns and compare outlets, while eight indicated a preference for using the system to analyze political news. Participants particularly valued the time-series and pie-chart visualizations, time-period filtering, outlet comparison, and article-level tone information. Suggested improvements included clearer explanations of EST/OPP tone and visualization labels, simultaneous EST/OPP analysis, and outlet ownership.

IV. Conclusion

We present *IndiaMediaLens*, an interactive system for real-time analysis of aggregate political reporting preferences of Indian digital news outlets through longitudinal and current coverage. By modeling political tone as targeted sentiment toward specific political entities, the system provides users with interpretable evidence of political reporting patterns. Our preliminary evaluation indicates the system's usefulness around systematic evaluation of news reporting preferences. As a pilot for media monitoring in India, IndiaMediaLens provides a foundation for incorporating broader news sources and analytical capabilities.

References

[1] S. Chakraborty, P. Goyal, and A. Mukherjee, "(im) balance in the representation of news? an extensive study on a decade long dataset from india," in *International Conference on Social Informatics*. Springer, 2022, pp. 347–369.

[2] A. F. Fonseca, S. Bandyopadhyay, J. Louçã, and J. A. Manjaly, "Caste in the news: A computational analysis of indian newspapers," *Social Media+ Society*, vol. 5, no. 4, p. 2056305119896057, 2019.

[3] AllSides Technologies, "AllSides," https://www.allsides.com/unbiased-balanced-news, 2012, [Online; accessed 18-August-2026].

[4] A. Panda, A. Gonawela, S. Acharyya, D. Mishra, M. Mohapatra, R. Chandrasekaran, and J. Pal, "Nivaduck-a scalable pipeline to build a database of political twitter handles for india and the united states," in *International Conference on Social Media and Society*, 2020, pp. 200–209.

[5] Reuters Institute, "Digital News Report 2025," https://reutersinstitute.politics.ox.ac.uk/digital-news-report/2025, 2025, [Online; accessed 18-August-2026].

[6] H. Roberts, R. Bhargava, L. Valiukas, D. Jen, M. M. Malik, C. S. Bishop, E. B. Ndulue, A. Dave, J. Clark, B. Etling *et al.*, "Media cloud: Massive open source collection of global news on the open web," in *Proceedings of the International AAAI Conference on Web and Social Media*, vol. 15, 2021, pp. 1034–1045.

[7] L. Ou-Yang, "Newspaper." [Online]. Available: https://pypi.org/project/newspaper3k/

[8] C. Goldberg, D. Burns, D. Molina, P. C. Jagani, and T. Fortner, "Python language bindings for selenium webdriver." [Online]. Available: https://pypi.org/project/selenium

[9] C. Scanlan, "Writing from the top down: Pros and cons of the inverted pyramid — poynter," https://www.poynter.org/reporting-editing/2003/writing-from-the-top-down-pros-and-cons-of-the-inverted-pyramid/, 2003, [Online; accessed 18-August-2026].

[10] S. Biswas, "Google News Decoder." [Online]. Available: https://pypi.org/project/googlenewsdecoder/

[11] Wikipedia contributors, "National democratic alliance — Wikipedia, the free encyclopedia," https://en.wikipedia.org/w/index.php?title=National_Democratic_Alliance&oldid=1369400205, 2026, [Online; accessed 18-August-2026].

[12] ——, "Indian national developmental inclusive alliance — Wikipedia, the free encyclopedia," https://en.wikipedia.org/w/index.php?title=Indian_National_Developmental_Inclusive_Alliance&oldid=1369651885, 2026, [Online; accessed 18-August-2026].

[13] M. Honnibal, I. Montani, S. Van Landeghem, A. Boyd *et al.*, "spacy: Industrial-strength natural language processing in python," 2020.

[14] twitterapi.io, "TwitterAPI.io," https://twitterapi.io/, [Online; accessed 18-August-2026].

[15] F. Hamborg, "NewsSentiment: easy-to-use, high-quality target-dependent sentiment classification for news articles." [Online]. Available: https://pypi.org/project/NewsSentiment/

[16] Zauberzeug GmbH, "NiceGUI." [Online]. Available: https://pypi.org/project/nicegui/